BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

Note: Choose the correct answer for each from the given options:

Q1. The sequence of the first 100 natural numbers which are divisible by "5" will

(a) G.P
(b) H.P
(c) A.P
(d) None of these

Q2. The sum of first n' odd natural are:

(a) n
(b) n^2
(c) n(n+1)
(d) n(n+1)/2

Q3. If O(AUB) = 52, O(A) = 47, O(B) = 39 then O(A∩B) = _____

(a) 33
(b) 86
(c) 32
(d) 139

Q4. The sum of the roots of (a - b) x^2 - (b - a) x - co = 0 is _____.

(a) 0
(b) 1
(c) -1
(d) a-b

Q5. The number of circular permutations of 4 beads arranged in a circle is

(a) 3
(b) 4
(c) 5
(d) 6

Q6. If ω be cube roots of unity then ω + ω^2 + ω^3 = _____

(a) -1
(b) 0
(c) 1
(d) 3

Q7. The factorial of positive integer 'n' is _____.

(a) n(n-1)..3,2,1
(b) n(n-2)!
(c) n(n-1)(n-2)
(d) all are true

Q8. 2 sin 30° cose 45° = _____

(a) 2/2
(b) 2/2
(c) 1
(d) None of these

Q9. 2π radians is equal to _____ angles

(a) 2 rights
(b) 3 rights
(c) 4 rights
(d) None of these

Q10. There are 10 balls of different colours. In _____ ways can select 7 if tgen si as to include one red ball.

(a) 82
(b) 84
(c) 85
(d) 86

Q11. sin 2θ = _____

(a) 2 sin θ
(b) 2 sin θ cos θ
(c) cos^2 θ
(d) None of these

Q12. If tan θ = a, then sin θ = _____.

(a) a/(a^2+1)
(b) a/(a^2+1)
(c) a/(a^2+1)
(d) None of these

Q13. The expansion of (2 a; a 2) will contain _____ terms.

(a) Single
(b) 6
(c) 5
(d) 7

Q14. If n+1C3 = 2, nC2 then n = _____

(a) 3
(b) 4
(c) 5
(d) 6

Q15. Additive inverses of 3 + 5i is _____

(a) 3 + 5i
(b) 3 - 5i
(c) -3 + 5i
(d) -5i

Q16. A complete revolution is measurement of an arc of circle is equal to its _____

(a) radius
(b) diameter
(c) circumference
(d) None of these

Q17. Solution of six x + cosx = 0 is _____

(a) {2π/3, nπ}
(b) {3π/3, nπ}
(c) {π/4, nπ}
(d) None of these

Q18. Which of the following is the roots of sec x = 2 :

(a) π/4, π/4
(b) π/4, 3π/4
(c) 2π/4, 5π/4
(d) π/3, π/4

Q19. (-3 + i)/(-3i) = _____.

(a) 3/10
(b) 10/3
(c) 10/3
(d) None of these

Q20. If nP2 = 30, then n = _____

(a) 4
(b) 5
(c) 6
(d) 12

Q21. If A∩B = B then which one is true:

(a) A∩B = B
(b) A ⊂ B
(c) B ⊂ A
(d) None of these

Q22. If for a quadratic equation the discriminate b^2 - 4ac < 0, then roots are _____

(a) Complex and un-equal
(b) Equal and real
(c) Real and un-equal
(d) None of these

Q23. If Z1 = 2 - 3i and Z2 = 2 - 7i then Re ($\bar{Z}$1 + Z2) is _____

(a) -2
(b) 2
(c) -4
(d) 4

Q24. The general term in expansion of (a + b)^m is, form m > r:

(a) T_r+1 = Cr^m b_r
(b) T_r+1 = Cr^m a^(m-r) b^r
(c) T_r+1 Cr^m a^r b^m
(d) None of these

Section-B Short Answer

> Note: Answer any SIX of the following questions.

Q2. If U = {x/x ∈ Z, -1 ≤ x ≤ 5} A = {y/y ∈ W, 1 ≤ y < 4}, B = {z/z ∈ N, 2 < z [illegible in source scan] then verify that B - A = B ∩ A'.

Q3. Find the multiplicative inverse of -1/2 + 1/2 i

Q4. If the sum of 8 terms of an arithmetic's sequence is 64 and that of 19 terms is 361. Find the sum of 29 terms.

Q5. Prove the proposition 1 + 5 + 9 + …. (4n - 3) = n (2n - 1) by [illegible in source scan] Mathematical induction.

Q6. Find the equation whose roots are reciprocal the roots of x^2 - 6x + 8 = 0

Q7. If cos α = 1/2, cos β = 3/2 and α, β are in first quadrant then find cos (α - [illegible in source scan]

Q8. Find the value of "n" when 2n+1P_{n-1} : 2n-1P_a = 3:5.

Q9. Write in simplified form, the term independent of x in (x - 1/2)^[illegible in source scan]

Q10. Show that the roots of the following equation is rational. (a - b)x^2 + (b + c - a)x - c = 0, a, b, c ∈ Q, a ≠ b.

Section-C (Descriptive Answer)

> Note: Answer any TWO of the following questions.

Q11. (a) Show that: sin 2θ / (1 - cos 2θ) = cot θ Solve: 2 sin x cos x - sin x = 0

Q12. (a) Solve the equation: 2(y^2 + 1/y^2) - 6(y + 1/y) = 4 (b) Find the smallest angle in triangle ABC with a = 4cm, b = 2 cm, c = 2.5cm

Q13. (a) Find the sum of 20 terms of an AP whose 4th term is 8 and 7th term is 13. (b) Prove that: (1 - cos θ)/sin θ = sin θ/(1 - cos θ)